

C Tasks

Week 5

Q1: 1-Write function to find the greatest common divisor (GCD) of two numbers using Euclidean Algorithm.

Here is more about GCD: https://en.wikipedia.org/wiki/Euclidean_algorithm

2-Write function that takes two strings represent integer numbers and returns the maximum number of them numerically and the maximum of them lexicographically the strings don't have zeros on the left. Hint (you may use call by reference) example input "12" and "4" output "12" and "4" where "4" is the maximum lexicographically and "12" is the maximum numerically

3- Implement the Bubble Sort algorithm in C and analyze its time complexity and space complexity.

Q2: 1- Write a C function to implement linear search on an array and state its time and space complexity.

2- Implement the insertion and selection Sort algorithm in C and analyze their time complexity and space complexity.

3- write a function to get the nth number in Fibonacci series (memory and time must be as minimum as possible). (The proper solution using recursion will get higher degrees).

Q3: 1-Write a C program to implement the binary search algorithm to check if element is inside the array or not and state its time and space complexity. And output number of elements is bigger than given number and less than given number. (array is sorted)

2-Write a C program to implement the jump search algorithm to check if element is inside the array or not and state its time and space complexity. (array is sorted)

3-write a C program that predicts a number the user choose by asking questions to user and get answer as 'N' or 'n' for(No) and 'y' or 'Y' for(Yes) The number will be in range from 1 to 1000 in number of questions less than or equal to 10.

Output samples of Q3.3:

```
"C:\Users\ALKING\Desktop\C Basics\bin\Debug\C Basics.exe"
Q1 :is your number bigger than or equal to 500 ? n
Q2 :is your number bigger than or equal to 250 ? y
Q3 :is your number bigger than or equal to 375 ? y
Q4 :is your number bigger than or equal to 437 ? y
Q5 :is your number bigger than or equal to 468 ? n
Q6 :is your number bigger than or equal to 452 ? n
Q7 :is your number bigger than or equal to 444 ? n
Q8 :is your number bigger than or equal to 440 ? n
Q9 :is your number bigger than or equal to 438 ? y
Q10 :is your number bigger than or equal to 439 ? n
your number is 438
Process returned 0 (0x0)    execution time : 40.609 s
Press any key to continue.
```

```
"C:\Users\ALKING\Desktop\C Basics\bin\Debug\C Basics.exe"
Q1 :is your number bigger than or equal to 500 ? n
Q2 :is your number bigger than or equal to 250 ? n
Q3 :is your number bigger than or equal to 125 ? n
Q4 :is your number bigger than or equal to 62 ? n
Q5 :is your number bigger than or equal to 31 ? n
Q6 :is your number bigger than or equal to 15 ? n
Q7 :is your number bigger than or equal to 7 ? n
Q8 :is your number bigger than or equal to 3 ? n
Q9 :is your number bigger than or equal to 1 ? y
Q10 :is your number bigger than or equal to 2 ? n
your number is 1
Process returned 0 (0x0)    execution time : 14.095 s
Press any key to continue.
```

```
"C:\Users\ALKING\Desktop\C Basics\bin\Debug\C Basics.exe"
Q1 :is your number bigger than or equal to 500 ? y
Q2 :is your number bigger than or equal to 750 ? y
Q3 :is your number bigger than or equal to 875 ? y
Q4 :is your number bigger than or equal to 938 ? y
Q5 :is your number bigger than or equal to 969 ? y
Q6 :is your number bigger than or equal to 985 ? y
Q7 :is your number bigger than or equal to 993 ? y
Q8 :is your number bigger than or equal to 997 ? y
Q9 :is your number bigger than or equal to 999 ? y
Q10 :is your number bigger than or equal to 1000 ? y
your number is 1000
Process returned 0 (0x0)    execution time : 15.270 s
Press any key to continue.
```

Q4: 1-write a C program that predicts a number the user choose by asking questions to user if number is in range from num1 to num2 inclusive and get answer as '0' if is under the range and as '1' if is in range and as '2' if is upper the range The number will be in range from 1 to 1000 in number of questions less than or equal to 7. Hint (think to make the search range as three parts).

2- sort a 2d array of points (x and y) passed on the distance between them and specified point (solution must not be more than $O(n^2)$).

Here is how to calculate the distance :

<https://www.mathsisfun.com/algebra/distance-2-points.html>

Q5: 1-implement merge sort of an array and state only its time complexity.

Hint (you will use recursion so you must study it as much as possible to answer questions)

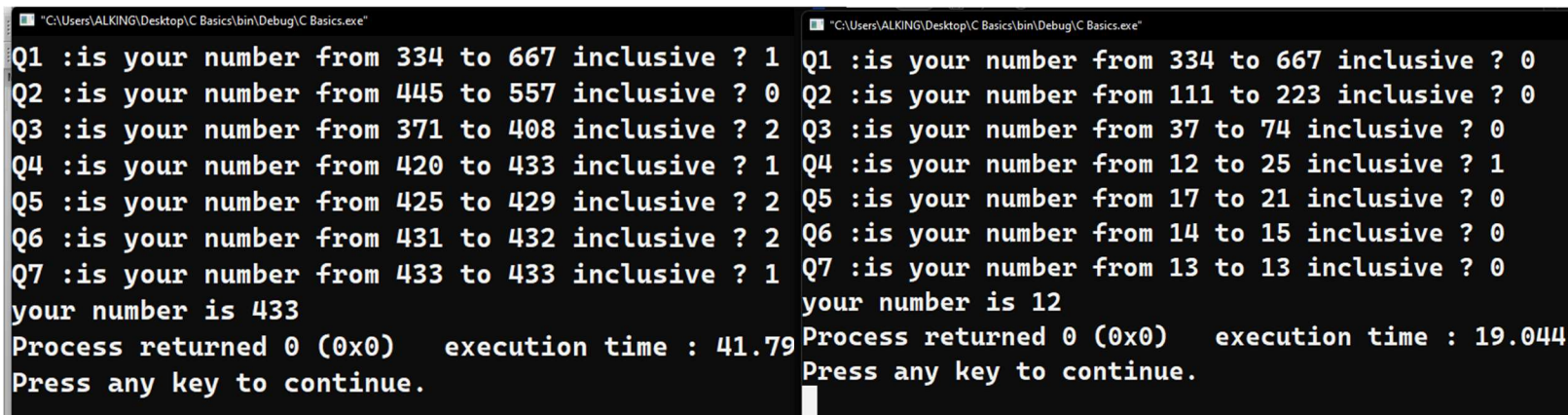
Here is more about merge sort:

<https://youtu.be/LeWuki7AQLo?si=HVAI7IJ4HvateXOq>

2- solve Q4.2 using merge sort and it (must not be $O(n^2)$).

3- Implement binary search to count the occurrences of a number in a sorted array.

Output samples of Q4.1



```
"C:\Users\ALKING\Desktop\C Basics\bin\Debug\C Basics.exe"
Q1 :is your number from 334 to 667 inclusive ? 1
Q2 :is your number from 445 to 557 inclusive ? 0
Q3 :is your number from 371 to 408 inclusive ? 2
Q4 :is your number from 420 to 433 inclusive ? 1
Q5 :is your number from 425 to 429 inclusive ? 2
Q6 :is your number from 431 to 432 inclusive ? 2
Q7 :is your number from 433 to 433 inclusive ? 1
your number is 433
Process returned 0 (0x0)   execution time : 41.79
Press any key to continue.

"C:\Users\ALKING\Desktop\C Basics\bin\Debug\C Basics.exe"
Q1 :is your number from 334 to 667 inclusive ? 0
Q2 :is your number from 111 to 223 inclusive ? 0
Q3 :is your number from 37 to 74 inclusive ? 0
Q4 :is your number from 12 to 25 inclusive ? 1
Q5 :is your number from 17 to 21 inclusive ? 0
Q6 :is your number from 14 to 15 inclusive ? 0
Q7 :is your number from 13 to 13 inclusive ? 0
your number is 12
Process returned 0 (0x0)   execution time : 19.044
Press any key to continue.
```

Bonus (must solve two of these four questions)

Q1: Given a structure array of student names and scores, sort them alphabetically and if two students have the same name sort them passed on their score.(solution should take $O(n \log n)$ hint (you may use merge sort that you used before)

Q2: Find the missing number in an array containing n unique numbers from 1 to n+1 hint (you may use sorting and try to solve it without sorting) (solution without sorting will get higher points) .

Q3: Given an array of numbers, find their mex ("m inimum ex cluded value"). (solution must be with $O(n \log n)$). It is guaranteed that Numbers in array isn't negative .

Here Is more about mex: [https://en.wikipedia.org/wiki/Mex_\(mathematics\)](https://en.wikipedia.org/wiki/Mex_(mathematics))

Q4: Given an array of n elements, find two numbers that sum up to a given number x . (you may use merge sort and binary search) (solution must be with $O(n \log n)$).

Output samples of bonus Q3:

```
"C:\Users\ALKING\Desktop\C Basics\bin\Debug\C Basics.exe"  
enter the number of elements in array10  
enter elements of the array 7 12 4 11 12 1 2 3 5 8  
the mex of the array is 0  
Process returned 0 (0x0)    execution time : 38.877 s  
Press any key to continue.
```

```
"C:\Users\ALKING\Desktop\C Basics\bin\Debug\C Basics.exe"  
enter the number of elements in array10  
enter elements of the array 5 3 4 2 0 1 7 12 9 10  
the mex of the array is 6  
Process returned 0 (0x0)    execution time : 34.194 s  
Press any key to continue.
```